

Bowen Tang

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Personal Information

- **Date of birth:** February 19, 2002
- **Place of birth:** Chongqing, China
- **Citizenship:** Chinese
- **Preferred pronouns:** He/Him/His
- **Website:** tbwtop.github.io

Education

- Peking University**, Ph. D. in Pure Mathematics, Beijing, China 2024-2029 (Expected)
- Advisor: Professor Yi Xie
- Peking University**, B. S. in Mathematics and Applied Mathematics, Beijing, China 2020-2024
- Undergraduate thesis: Yang-Mills Equations and the Topology of Four-Manifolds
 - Thesis advisor: Professor Yi Xie
 - GPA: 3.72/4

Research Interests

I am interested in low-dimensional topology. Some specific topics are gauge theory, instanton Floer homology, Khovanov homology, invariants of knots and links, and exotic four-manifolds.

Publications and Preprints

1. **A detection result of $SO(3)$ representation varieties on links in $S^1 \times S^2$**
Upcoming

Talks

- **Handle-addition formulas for skein lasagna modules**, *Seminar on Skein Lasagna Modules at Peking University*, Beijing, China, May 12, 2026.

Awards and Honors

- Outstanding Graduate (Beijing) 2024
- Bronze Medal in S.-T. Yau College Students Mathematics Contest 2022

Teaching Experience

- Peking University**
- Advanced Mathematics (B) I** 2026 Fall
Teaching Assistant. Lecturer: Professor Jianchun Chu
- Homology Theory** (Enhanced Program for Graduate Study at BICMR) 2026 Spring
Teaching Assistant. Lecturer: Professor Yi Xie
- Linear Algebra (B)** 2025 Fall
Teaching Assistant. Lecturer: Professor Guchuan Li
- Fiber Bundles and Characteristic Classes** 2025 Spring
Teaching Assistant. Lecturer: Professor Zhiqiang Bao
- Geometry** 2024 Fall
Teaching Assistant. Lecturer: Professor Yi Xie
- Advanced Mathematics (A) I** 2023 Fall
Teaching Assistant. Lecturer: Professor Jinbiao Wu

Service

I serve as the Deputy Secretary of the CPC Branch for Pure Mathematics PhD Students at School of Mathematical Sciences, Peking University.

I have been actively involved in undergraduate admissions at Peking University.

Language

Mandarin (native), **English** (fluent)